

Assignment 5: Data Visualization

Sloane Hauck

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Visualization

Directions

1. Rename this file `<FirstLast>_A05_DataVisualization.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up your session

1. Set up your session. Load the tidyverse, here & cowplot packages, and verify your home directory. Read in the NTL-LTER processed data files for nutrients and chemistry/physics for Peter and Paul Lakes (use the tidy NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv version in the Processed_KEY folder) and the processed data file for the Niwot Ridge litter dataset (use the NEON_NIWO_Litter_mass_trap_Processed.csv version, again from the Processed_KEY folder).
2. Make sure R is reading dates as date format; if not change the format to date.

#1

```
library(tidyverse);library(here);library(cowplot);library(ggplot2);library(ggthemes)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    4.0.0      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
## here() starts at /home/guest/EDE_Fall2025
##
##
## Attaching package: 'cowplot'
##
##
## The following object is masked from 'package:lubridate':
##
##     stamp
##
##
## Attaching package: 'ggthemes'
##
##
## The following object is masked from 'package:cowplot':
##
##     theme_map
```

```
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
Lake_Chemistry <- read.csv(
  file = here("Data/Processed_KEY/NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv"),
  stringsAsFactors = TRUE
)
```

```
Litter_Mass <- read.csv(
  file = here("Data/Processed_KEY/NEON_NIWO_Litter_mass_trap_Processed.csv"),
  stringsAsFactors = TRUE
)
```

```
#2
class(Lake_Chemistry$sampledate)
```

```
## [1] "factor"
```

```
Lake_Chemistry$sampledate <- ymd(Lake_Chemistry$sampledate)
class(Litter_Mass$collectDate)
```

```
## [1] "factor"
```

```
Litter_Mass$collectDate <- ymd(Litter_Mass$collectDate)
```

Define your theme

3. Build a theme and set it as your default theme. Customize the look of at least two of the following:

- Plot background

- Plot title
- Axis labels
- Axis ticks/gridlines
- Legend

```
#3
my_theme <- theme_base() +
  theme(
    plot.title = element_text(size = 12,
                              color = "darkblue"),
    axis.title.x = element_text(size = 10,
                                color = "darkblue"),
    axis.title.y = element_text(size = 10,
                                color = "darkblue"),
    axis.ticks = element_line(color = "black"),
    plot.background = element_rect(color = "black",
                                   fill = "lightgrey"),
    complete = TRUE
  )
```

Create graphs

For numbers 4-7, create ggplot graphs and adjust aesthetics to follow best practices for data visualization. Ensure your theme, color palettes, axes, and additional aesthetics are edited accordingly.

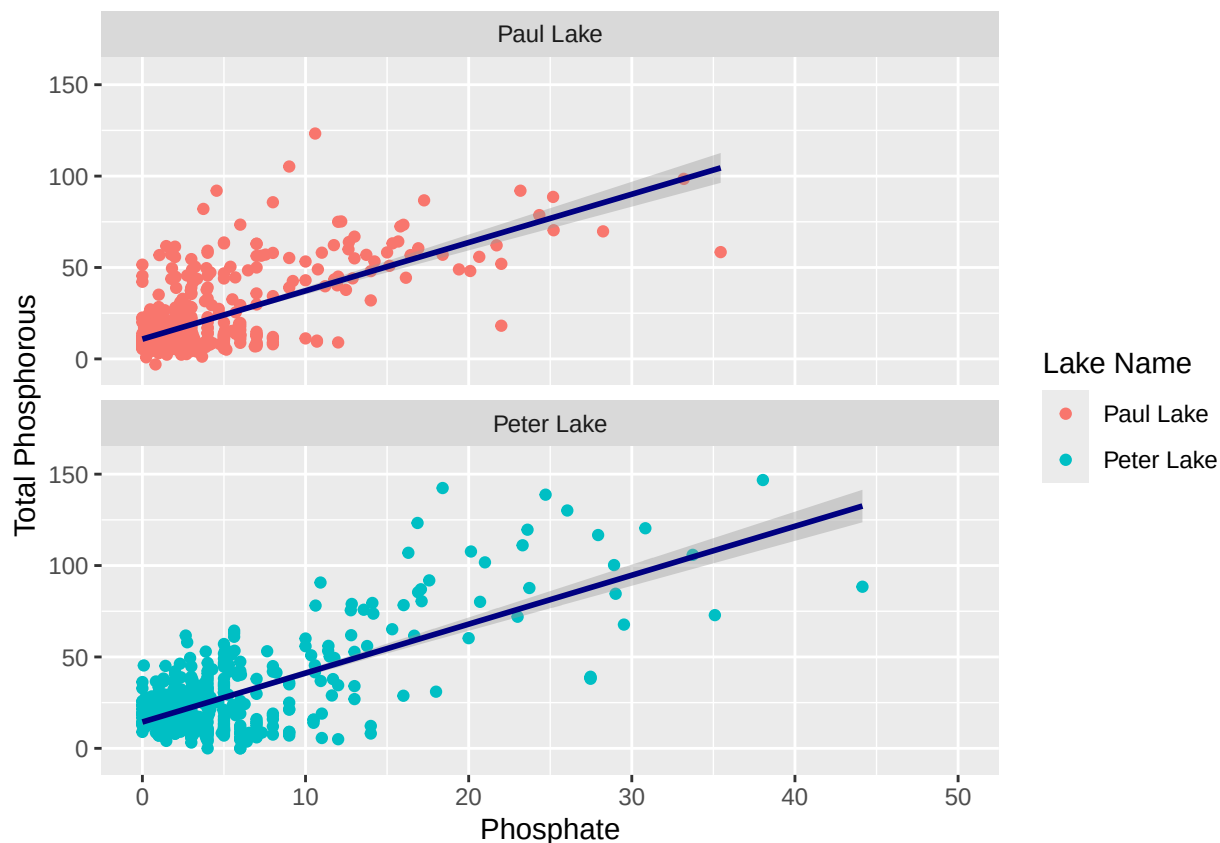
4. [NTL-LTER] Plot total phosphorus (tp_ug) by phosphate (po4), with separate aesthetics for Peter and Paul lakes. Add line(s) of best fit using the `lm` method. Adjust your axes to hide extreme values (hint: change the limits using `xlim()` and/or `ylim()`).

```
#4
ggplot(Lake_Chemistry, aes(x=po4, y=tp_ug, color = lakename)) +
  geom_point() +
  facet_wrap(facets = vars(lakename),
            nrow = 2) +
  ylab("Total Phosphorous") +
  xlab("Phosphate") +
  xlim(0,50) +
  labs(color = "Lake Name" ) +
  geom_smooth(method = lm, color = "navy") +
  theme(my_theme)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 21947 rows containing non-finite outside the scale range
## ('stat_smooth()').
```

```
## Warning: Removed 21947 rows containing missing values or values outside the scale range
## ('geom_point()').
```



5. [NTL-LTER] Make three separate boxplots of (a) temperature, (b) TP, and (c) TN, with month as the x axis and lake as a color aesthetic. Then, create a cowplot that combines the three graphs. Make sure that only one legend is present and that graph axes are aligned.

Tips: * Recall the discussion on factors in the lab section as it may be helpful here. * Setting an axis title in your theme to `element_blank()` removes the axis title (useful when multiple, aligned plots use the same axis values) * Setting a legend's position to "none" will remove the legend from a plot. * Individual plots can have different sizes when combined using `cowplot`.

```
#5
Temp <- Lake_Chemistry %>%
  ggplot(aes(x = factor(month, levels = 1:12, labels = month.abb),
             y = temperature_C, fill = lakename)) +
  geom_boxplot() +
  scale_fill_brewer(palette = "Pastell1") +
  # scale_x_discrete(drop = FALSE) +
  xlab("Month") +
  ylab("Temperature (C)") +
  theme(my_theme, legend.position = "None")

Total_Phosphorous <- Lake_Chemistry %>%
  ggplot(aes(x = factor(month, levels = 1:12, labels = month.abb),
             y = tp_ug, fill = lakename)) +
  geom_boxplot() +
  scale_fill_brewer(palette = "Pastell1") +
```

```

# scale_x_discrete(drop = FALSE) +
  xlab("Month") +
  ylab("Total Phosphorous (ug)") +
  theme(my_theme, legend.position = "None")

Total_Nitrogen <- Lake_Chemistry %>%
  ggplot(aes(x = factor(month, levels = 1:12, labels = month.abb),
             y = tn_ug, fill = lakename)) +
  geom_boxplot() +
  scale_fill_brewer(palette = "Pastell1") +
  # scale_x_discrete(drop = FALSE) +
  xlab("Month") +
  ylab("Total Nitrogen (ug)") +
  theme(my_theme) +
  labs(fill = "Lake Name")

plot_grid(Temp, Total_Nitrogen, Total_Phosphorous, nrow = 3,
          align = 'h', rel_heights = c(1, 1))

```

```

## Warning: Removed 3566 rows containing non-finite outside the scale range
## ('stat_boxplot()').

```

```

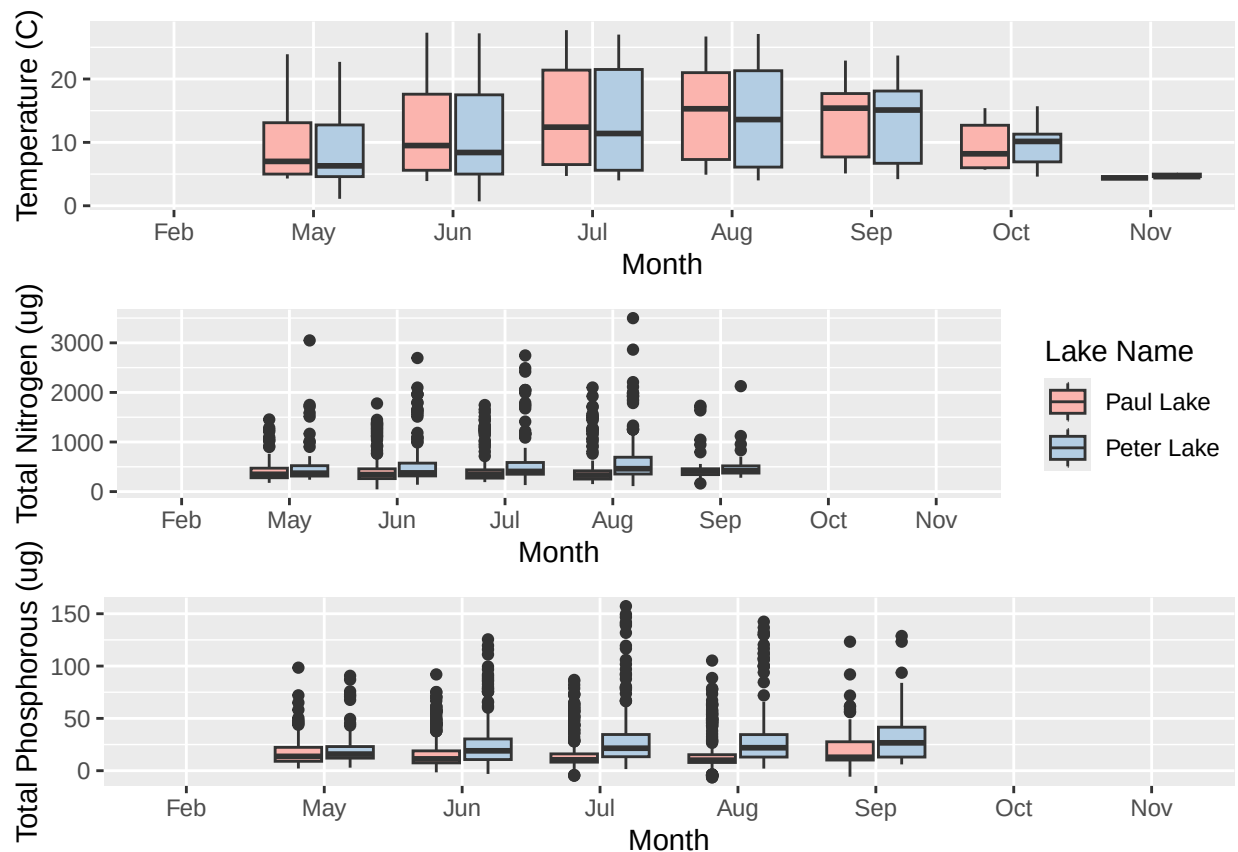
## Warning: Removed 21583 rows containing non-finite outside the scale range
## ('stat_boxplot()').

```

```

## Warning: Removed 20729 rows containing non-finite outside the scale range
## ('stat_boxplot()').

```



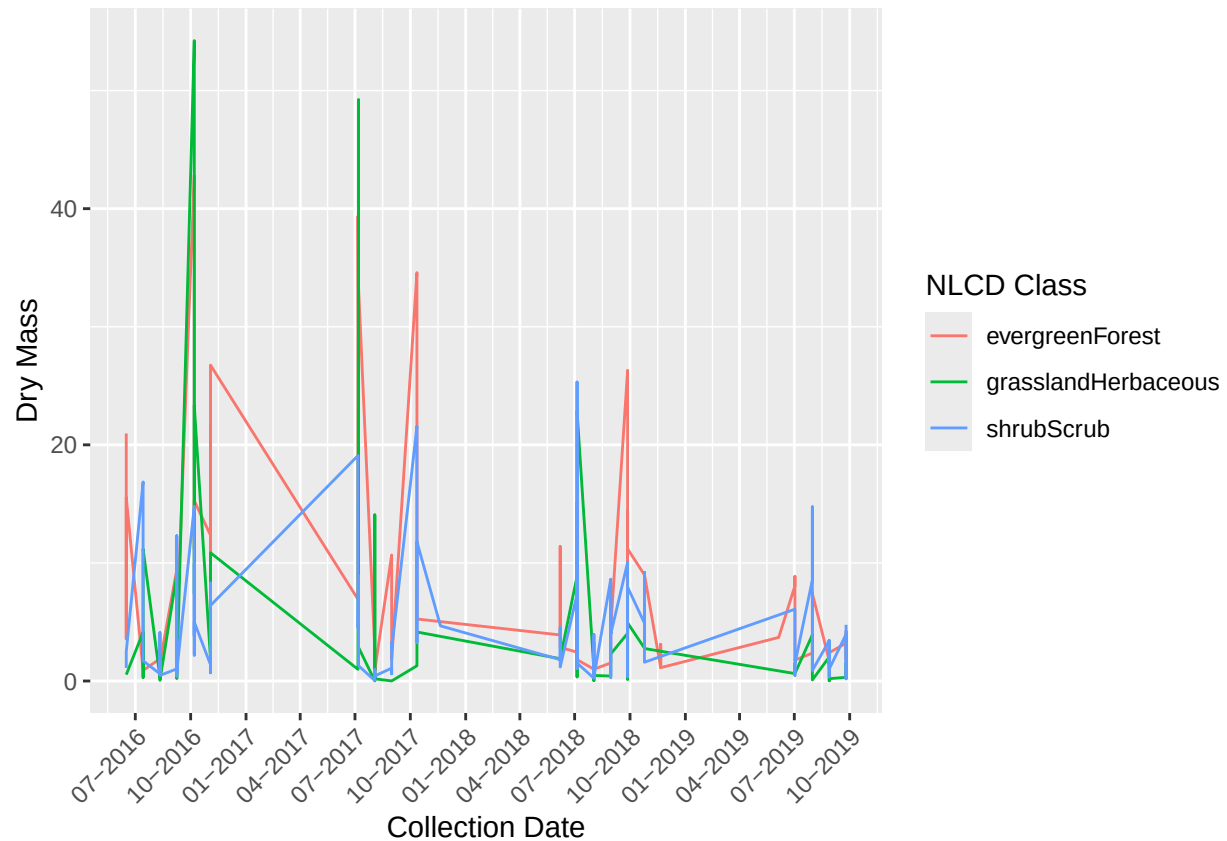
Question: What do you observe about the variables of interest over seasons and between lakes?

Answer: Temperature is higher in the summer months, as would be expected. Additionally, the nutrient levels are also slightly higher by the end of the summer/early autumn.

6. [Niwot Ridge] Plot a subset of the litter dataset by displaying only the “Needles” functional group. Plot the dry mass of needle litter by date and separate by NLCD class with a color aesthetic. (no need to adjust the name of each land use)
7. [Niwot Ridge] Now, plot the same plot but with NLCD classes separated into three facets rather than separated by color.

```
#6
Needles <- Litter_Mass %>%
  filter(functionalGroup == "Needles") %>%
  ggplot(aes(x = collectDate, y = dryMass, color = nlcdClass)) +
  geom_line() +
  theme(my_theme, axis.text.x = element_text(angle = 45, hjust = 1)) +
  scale_x_date(date_breaks = "3 month", date_labels = "%m-%Y") +
  xlab("Collection Date") +
  ylab("Dry Mass") +
  labs(color = "NLCD Class")

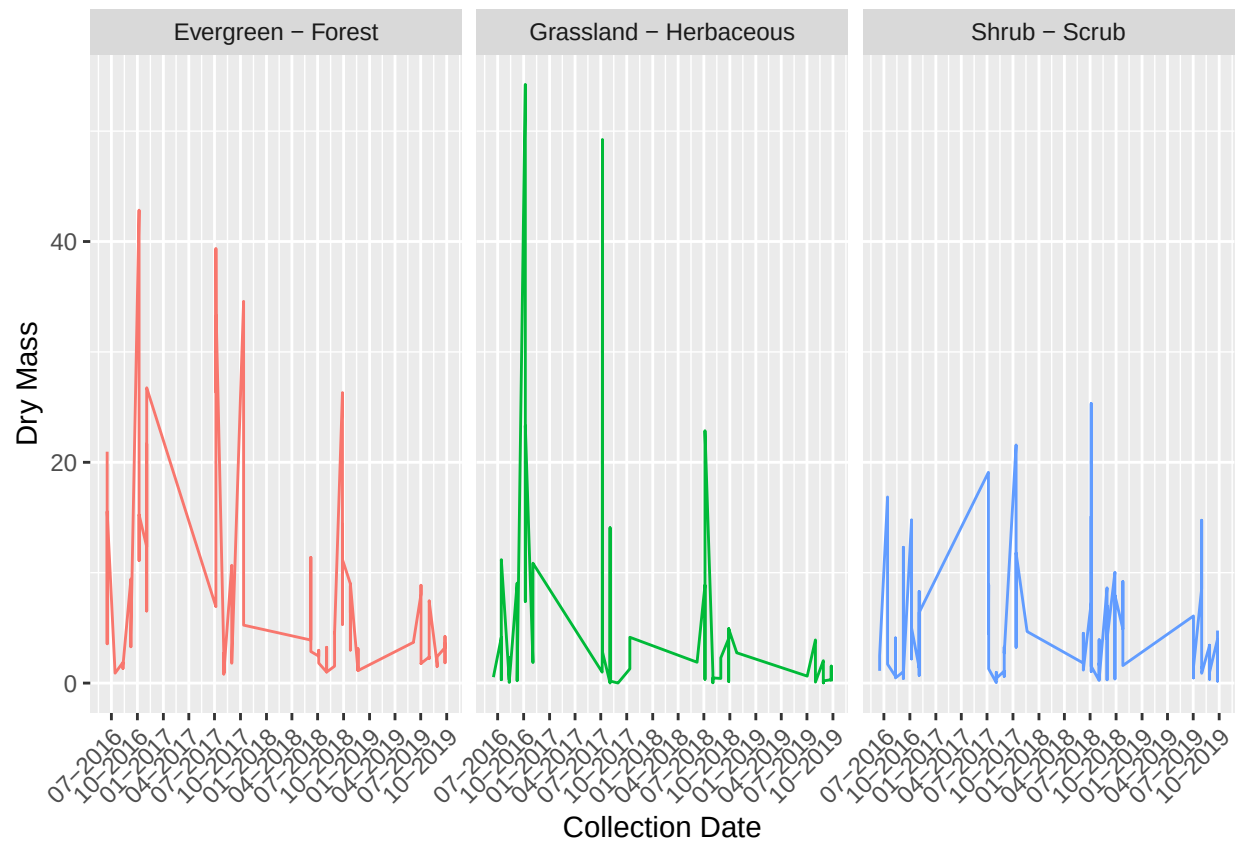
Needles
```



#7

```
Needles_Separate <- Litter_Mass %>%
  filter(functionalGroup == "Needles") %>%
  mutate(nlcdClass = case_when(
    nlcdClass %in% "evergreenForest" ~ "Evergreen - Forest",
    nlcdClass %in% "grasslandHerbaceous" ~ "Grassland - Herbaceous",
    nlcdClass %in% "shrubScrub" ~ "Shrub - Scrub",
    TRUE ~ nlcdClass
  )) %>%
  ggplot(aes(x = collectDate, y = dryMass, color = nlcdClass)) +
  geom_line() +
  facet_wrap(vars(nlcdClass)) +
  theme(my_theme, axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position = "None") +
  scale_x_date(date_breaks = "3 month", date_labels = "%m-%Y") +
  xlab("Collection Date") +
  ylab("Dry Mass") +
  labs(color = "NLCD Class")
```

Needles_Separate



Question: Which of these plots (6 vs. 7) do you think is more effective, and why?

Answer: I think that 7 is more effective because it is easier to see trends over time and by having the graphs sit next to each other, you can still see what overlapping trends exist.